

Retrieval-Grounded Arabic Fact-Checking: Extending ARAFA with Evidence Retrieval, Model Ensembling, and Robustness Auditing

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Abstract

Automatic fact-checking in Arabic remains constrained by the scarcity of large-scale resources. ARAFA, a recently introduced LLM-generated Arabic fact-checking dataset of 181,976 claim-evidence pairs, addresses this scarcity but frames fact-checking purely as *verification*: every claim in ARAFA is supplied with its gold evidence already attached, so the models trained and evaluated on it are never required to find that evidence themselves. Real-world fact-checking, by contrast, is a *retrieve-then-verify* process. This thesis proposes to close that gap by (1) building and evaluating a retrieval layer over Arabic Wikipedia that recovers the evidence ARAFA claims were originally generated from, (2) re-evaluating existing verification models under retrieved rather than gold evidence to quantify the resulting performance gap, (3) developing an ensemble/routing model that combines the complementary strengths of encoder-based and generative verifiers observed in the original ARAFA study, and (4) conducting a systematic audit to confirm that performance gains are attributable to genuine evidence use rather than to memorization or shortcut learning. The resulting system, evaluation protocol, and released retrieval index are intended to serve as a reusable benchmark for future work on Arabic fact-checking and Arabic retrieval-augmented generation more broadly.

1 Introduction

Automatic fact-checking has advanced substantially for high-resource languages such as English, driven largely by benchmark datasets such as FEVER [2], which frame the task as retrieving evidence from a large corpus and then verifying a claim against it. Arabic fact-checking has lagged behind, both in dataset scale and in task realism: existing Arabic resources such as AraFacts [4] and AraStance [5] are small and domain-limited, typically drawn from news or social media.

ARAFA [1] was recently introduced to address the scale problem, using large language models (GPT-4o for generation and refutation, Claude Sonnet 3.5 for validation) to automatically construct 181,976 Arabic claim-evidence pairs from Wikipedia, labeled as *supported*, *refuted*, or *not enough information* (NEI). Four transformer-based models were fine-tuned and evaluated on ARAFA, reaching a best macro-F1 of 77%.

However, ARAFA addresses only half of the fact-checking problem. Because each claim is generated directly from the same Wikipedia passage that serves as its evidence, every model in the original study is evaluated under an *oracle* setting: the correct evidence is always handed to the model. No retrieval step is built or tested, despite “evidence retrieval” appearing among the paper’s own keywords. This is a

meaningful gap, since in practice a fact-checking system must first locate relevant evidence from a large, unstructured corpus before it can be verified against a claim – and retrieval errors are a primary source of failure in real-world fact-checking pipelines.

A second, related concern motivates part of this proposal: with modern large language models increasingly shipping built-in web search and retrieval, it is reasonable to ask what a purpose-built retrieval layer contributes that a general-purpose LLM does not already provide. This thesis treats that question directly rather than assuming it away. Unlike black-box LLM search, a controlled retrieval layer built over a known corpus with known ground-truth evidence allows retrieval quality to be measured independently of verification quality (via $\text{Recall}@k$ and Mean Reciprocal Rank), allows the two stages to be ablated separately, and allows a rigorous test of whether a verifier’s predictions are actually grounded in retrieved evidence or are instead a product of parametric memorization. This attribution question – is the retrieved evidence doing the work, or is the underlying language model doing it regardless of what evidence it is given – is addressed explicitly in Objective 4 below.

2 Related Work

The FEVER dataset [2] established the standard retrieve-then-verify formulation for automatic fact-checking, pairing claims with Wikipedia evidence retrieved from a large corpus rather than supplied directly. Extensions such as FEVEROUS [3] broadened this to structured and unstructured evidence jointly. Retrieval-augmented approaches more broadly [7] have shown that grounding a language model’s output in retrieved passages improves factual accuracy and reduces hallucination, provided the retrieval step is reliable.

For Arabic specifically, AraFacts [4] and AraStance [5] provide naturally occurring claims but at limited scale, while AuRED [6] focuses on rumor verification using evidence from social media authorities. None of these resources pairs a large-scale claim set with a matching retrieval corpus and ground-truth evidence locations in the way ARAFA’s provenance data makes possible. On the modeling side, multilingual dense retrievers such as BGE-M3 [8] and multilingual E5 [9], alongside classical lexical retrieval (BM25) [10], provide strong off-the-shelf starting points for building an Arabic retrieval layer without training a retriever from scratch.

Finally, work on shortcut learning and spurious correlations in NLI and fact-verification models [11] demonstrates that models can achieve high accuracy on fact-verification benchmarks using surface artifacts in claims alone, without using the evidence at all – a concern directly relevant to ARAFA’s refutation task, which is constructed using a small, fixed set of mutation strategies.

3 Objectives

Objective 1: Build and Evaluate a Retrieval Layer over Arabic Wikipedia

- Recover, for each ARAFA claim, its originating Wikipedia passage using the dataset’s preserved source-chunk and coreference metadata.
- Construct a passage-level index over Arabic Wikipedia using both lexical (BM25) and dense (multilingual embedding model) retrieval, and a hybrid combination of the two.
- Evaluate retrieval quality independently of verification, using $\text{Recall}@k$ ($k = 1, 5, 10$) and Mean Reciprocal Rank against the known gold passage for each claim.

Objective 2: Verify Claims under Retrieved (Non-Gold) Evidence

- Re-evaluate the encoder-based and generative verification models on the same claim set, but supplied with top- k retrieved passages instead of gold evidence.
- Report and analyze the performance gap between the oracle setting (gold evidence, as in the original ARAFA study) and the pipeline setting (retrieved evidence), to quantify the real-world cost of removing the oracle assumption.

Objective 3: Develop an Ensemble Verification Model

- Exploit the complementary error patterns observed in the original ARAFA study, where encoder-based models are strong on *supported* claims but weak on NEI, while generative models show the reverse pattern.
- Train a lightweight gating/meta-classifier that routes each claim-evidence pair to the more reliable model type, incorporating retrieval confidence as an additional signal.
- Evaluate the ensemble against the strongest single model under the pipeline (retrieved-evidence) setting, where the value of combining models is expected to be greatest.

Objective 4: Audit for Shortcut Learning and Verify Evidence Attribution

- Run a closed-book baseline (claim only, no evidence) to establish how much accuracy is achievable from memorized knowledge alone.
- Compare verification accuracy on claims where retrieval succeeded versus claims where it failed, to test whether the verifier’s correctness actually depends on receiving correct evidence.
- Run a perturbation test (substituting a random, topically unrelated passage) and, where feasible, a counterfactual test (evidence that directly contradicts the claim), to determine whether the verifier is grounded in the evidence it is given or is defaulting to parametric knowledge.
- Break down refutation accuracy by the five mutation strategies used to construct ARAFA’s refuted claims (temporal, quantitative, qualitative, relationship, contextual), to identify which categories are genuinely difficult versus easily detected surface artifacts.

4 Methodology

Data Collection and Processing

The ARAFA dataset (181,976 claim-evidence pairs, publicly available under a CC-BY license) will serve as the primary source of claims, gold evidence, and entity coreference annotations. A full Arabic Wikipedia dump will be downloaded, cleaned following the same preprocessing described in the original ARAFA paper (removal of templates, links, and category tags), and split into retrieval-sized passages (approximately 100–400 words), distinct from the larger generation-time chunks used to construct ARAFA. Each ARAFA claim’s originating passage will be identified and treated as ground truth for retrieval evaluation.

Retrieval Model Development

Passages will be indexed using FAISS for dense retrieval and a standard inverted-index implementation for BM25. Off-the-shelf multilingual embedding models (e.g., BGE-M3, multilingual-E5) will be used as a first baseline; if time permits, a bi-encoder retriever will be fine-tuned on claim→gold-passage pairs using in-batch and BM25-mined hard negatives. Hybrid retrieval (score fusion of BM25 and dense results) will be evaluated as an additional configuration, given known weaknesses of purely dense retrieval on morphologically rich languages.

Verification Model Integration

The encoder-based (AraBERTv2, AraModernBert) and generative (Llama-3.1-8B, Qwen2.5-7B via LoRA) verification approaches described in the original ARAFA study will be reused where feasible, either as released or retrained under comparable settings, and evaluated under both the oracle and retrieved-evidence conditions described in Objective 2.

Ensemble Development

A meta-classifier (e.g., logistic regression or a small feed-forward network) will be trained on top of the individual verifiers’ output distributions, retrieval confidence scores, and any other lightweight per-instance features, using a held-out portion of the data disjoint from the final test set.

Evaluation and Interpretability

Retrieval quality is measured with Recall@ k and MRR. Verification quality is measured with accuracy, macro-F1, and per-class F1 (supported / refuted / NEI), reported separately for the oracle and pipeline settings. Attribution and robustness are measured with closed-book accuracy, the retrieval-success conditioned accuracy gap, the perturbation-test accuracy delta, and per-mutation-strategy refutation accuracy.

Integration

The retrieval and verification components will be combined into a single pipeline accepting a raw Arabic claim as input and producing retrieved evidence, a verification label, and a confidence estimate as output, to demonstrate feasibility as an end-to-end system.

5 Data and Ethical Considerations

All data used in this thesis is derived from publicly available sources: the ARAFA dataset (Zenodo, CC-BY 4.0) and Arabic Wikipedia (public dump, no personal or sensitive data). No human-subject data, patient data, or personally identifiable information is involved. Any use of proprietary LLM APIs (e.g., for supplementary data generation) will follow the relevant providers’ terms of use and will be limited to non-sensitive, publicly sourced text. All code, the retrieval index, and evaluation scripts will be released publicly to support reproducibility, consistent with ARAFA’s own open-data policy.

6 Expected Results and Impact

This work is expected to produce:

- A benchmarked retrieval layer over Arabic Wikipedia, with Recall@ k and MRR results, usable as a reference point for future Arabic retrieval-augmented fact-checking work.
- A quantified comparison between oracle (gold-evidence) and realistic (retrieved-evidence) verification performance, clarifying how much of ARAFA’s reported 77% macro-F1 depends on the oracle assumption.
- An ensemble verification model that combines the complementary strengths of encoder-based and generative verifiers, evaluated specifically under realistic retrieval conditions.
- A documented evidence-attribution and shortcut-learning audit protocol, applicable beyond this thesis to other Arabic (and non-Arabic) retrieval-augmented systems.
- A publicly released retrieval index, evaluation code, and pipeline, intended for reuse by future researchers and practitioners building Arabic fact-checking or verification tools.

By turning ARAFA from a verification-only benchmark into a full retrieve-then-verify pipeline benchmark, this thesis aims to produce a more realistic and more reusable foundation for automatic Arabic fact-checking research.

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